No of Pages : 4 **Course Code : 23MX21**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004
SEMESTER EXAMINATIONS, JUNE 2025
MCA Semester: 2
23MX21 SOFTWARE ENGINEERING

Time: 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, (b) carries 7 marks each and (c) carries 15 marks each.
3. COCOMO table is permitted to the students**.**
4. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO2.	Qn.3	CO3.	Qn.4	CO4.

COs	At the end of learning this course, students will be able to
CO1	Analyse software development methodologies and life cycle models and select appropriate ones for the given problem and compute the project estimate.
CO2	Perform requirements gathering and prepare SRS for a given application as per IEEE format.
CO3	Apply UML for a sample OOAD application and prepare design documents as per IEEE
CO4	Select and apply appropriate testing methodologies for testing different types of software applications and analyse the test results

		Question	BTL
1.	a)	How the systematic approach will contribute to high quality and low cost software?	(L2)
	b) i)	What are the problems might a software development team face if they do not follow any life cycle models during development of a software product? (4)	(L3)
	ii)	Write a brief notes on “DevOps”. (3)	(L2)
	c)	Consider the following Organic project description, generate estimates for total effort, duration, number of people and the total cost required to develop using the COCOMO method. Progellis Technologies is developing an AI-powered financial analytics system for investment firms. The system must process large datasets, perform complex mathematical computations, and provide real-time insights. The project consists of three major modules with the following estimated sizes: Data Processing Engine: 30000 LOC, Machine Learning Module: 50 KLOC, Visualization Dashboard: 43000 LOC. Consider the following cost drivers: High - database size; very high - product complexity; very high -main storage constraint; high - execution time constraints; very high - programmer capability; very low - programming language experience. Use a=3.2, b=1.05, c=2.5 and d=0.38 Considering the following salary distribution to estimate the total man power cost for the development: ₹ 60000 per Analyst, ₹ 85000 per Designer, ₹ 120000 per Programmer, ₹ 55000 per Tester and a sum of ₹ 350000 for the Project Manager to lead the team for throughout the project.	(L3)
2.	a)	Is it true that, prototyping is a technique of building a software which helps in uncovering and capturing software requirements from client? Justify the answer	(L4)

b) i) A bank is developing a new online banking system, and the following requirements have been identified. Classify each requirement as functional or non-functional. (4) (L2)

1. The system shall allow users to register using an email address and a strong password.
2. The system shall provide an option for password reset via email verification.
3. The system shall lock the account after five consecutive failed login attempts and allow unlock via email verification.
4. The system shall enforce end-to-end encryption (TLS 1.2 or higher) for login credential transmission.
5. The system shall authenticate users within 2 seconds under normal load conditions.
6. The system shall automatically log out inactive users after 15 minutes of inactivity.

ii) During the requirement gathering for a payroll system, a stakeholder proposed the following requirement. Highlight any issues with the following requirement and rewrite the corrected version of the requirement. (L3)

"If an employee's salary exceeds a certain amount, the system should generate a notification to HR to review the salary details before processing the payroll" (3)

c) i) Design a neat USE CASE Model to depict the GOLD vending machine problem. It is explained as follows: (L6)

This machine is similar to ATM – it is a vending machine to buy gold coins; each coin weighs 1 gram. (5)

1. Customer apply for a GOLD card to access this machine. Gold card with sealed PIN is despatched to the customer. Customer uses this card for purchasing GOLD coin from the machine.
2. Customer scans the card in the machine; it validates the card and asks for the PIN.
3. Customer enters the PIN.
4. Machine displays the current rate of a coin.
5. Customer enter the number of coins interested to buy.
6. Machine calculates and displays the amount to be paid by the customer.
7. If the customer is interested to buy, he/she presses the "BUY" button or "CANCEL" button to cancel the transaction.
8. Machine asks for the mode of payment [Cash/ Card]
9. If the customer decided to pay CASH, machine accepts the same; checks the cash inserted in the machine; dispense the Gold coins along with the balance if any and a receipt.
10. Otherwise, machine reads the Card inserted by the customer; debit the amount from the card and dispense the Gold coins and a receipt.
11. Machine also handles any queries from the customer.

ii) Write a complete USE CASE Story to explain the above Gold Vending Machine problem (10) (L6)

3 a) "A design solution is simple and understandable, if it is modular". How to achieve this modularity? (L2)

b) Draw a **state chart** to depict the following scenario: (L6)

"A device is having a set of three radio buttons, b1, b2, and b3. The buttons are all initially off (un-pressed). Subsequently, if one button is pressed, it goes on and all the others go off"

c) i) Design a neat **sequence diagram** to depict the Coffee Machine Problem explained as below. (L6)

Coffee machine interface consists of a GUI, a coin slot for accepting change & returning change, a slot for delivering the cup of drink and selection buttons for coffee and tea.

The coffee machine can handle coins of the values 5 and 10, where 5 is the price for a cup of tea and 10 is the price for a cup of coffee. Internally, the coffee machine consists of a change verification device that counts money and makes change.

- If a coin with the value of 10 is inserted and the Coffee button is pressed - the customer receives a cup of coffee.
- If a coin with the value of 5 is inserted and the Tea button is pressed - the customer receives a cup of tea.
- If a coin with the value of 10 is inserted and the Tea button is pressed - the customer receives a cup of tea and the change.
- If a coin with the value of 5 is inserted and the Coffee button is pressed - the money is returned

[OR]

ii) Depict a complete **Activity Model** to narrate the procedure of withdrawing money from an ATM as explained below. (L6)

To withdraw cash, first valid customer identification is required. For this, the customer is prompted to insert his/her ATM card. If the card is found to be invalid, the card is ejected out from the machine with an appropriate message.

If the card is found to be a valid card, the customer is prompted to enter PIN. If the PIN is invalid, an error message is shown and the customer is prompted to enter the PIN again. If the customer enters incorrect PIN consecutively for three times, then the ATM card is seized and he/she is asked to contact the bank manager.

On the other hand, if the PIN is correct, then he/she is considered as a valid customer and is prompted to enter the amount he/she needs to withdraw.

If he enters an amount that is not multiples of Rs. 100, he/she is prompted to enter the amount again.

After he/she enters an amount that is multiples of 100, the cash is dispensed if sufficient amount is available in his/her account and the card is ejected with entered cash otherwise the card is ejected without any cash being dispensed along with a message display regarding insufficient fund position in his/her account.

4. a) What is the significance of Cyclomatic Complexity Constant? (L2)

b) i) "Exhaustive Testing is Impossible" – defend this with the help of an appropriate example. (4) (L3)

ii) What is the back ground of the "Defect Clustering" principle? (3) (L3)

c) i) 1) A bank has the following policy to pay Interest Rate for the deposits made by their customers based on period of deposit. If the deposit amount is Rs. 50,000 and above and for the period of deposit is 3 years and above, the interest rate is 12% of deposit amount. (L5)

For the same deposit, if the period of deposit is less than 3 years, the interest rate is 10%. If the deposit amount is less than Rs. 50,000, regardless of period of deposit, the interest rate is 8%.

Write a complete pseudo code to implement this logic and test the same using PATH Testing. (8)

2) Design suitable test cases using black box testing for a function "QuadraticSolver". It asks for three values a, b & c and solve the quadratic equation $ax^2+bx+c = 0$. Tabulate the test results in the standard format. (7) (L5)

[OR]

ii) A program determines the next date in the calendar. It reads the date in the format DD/MM/YYYY. The year lies in the range [1900, 2025]. Design test cases for this program using Boundary Value Analysis and Robust testing methods. Tabulate the test cases in the following order: BVA based and then Robust based testing. (L5)

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/RL